

5DGSPTSBL: Geometric Fast-Slow Stable Boundary Layer Model

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2026-08-02

Abstract

We present a five-dimensional geometric singular perturbation model for the stable boundary layer with state variables $(\tilde{e}, q_\theta, S, T_s, T_g)$. The model combines desingularized turbulence dynamics, slaved thermodynamic closures, and slow thermal coupling to support fold-locus diagnostics and data-driven parameter identification.

1 Closures and Smooth Regularization

To maintain C^r -smoothness near neutral stratification and support robust implicit integration with autodiff Jacobians, the stratification closure uses a hyperbolic embedding of $\max(0, x)$:

$$\text{smooth_max}(x; \epsilon) = \frac{x + \sqrt{x^2 + \epsilon^2}}{2}. \quad (1)$$

With $x = (\theta_0 - T_s)/h_{\text{sbl}}(\tilde{e})$, the closure becomes

$$\theta_z(T_s, \tilde{e}) = \text{smooth_max}\left(\frac{\theta_0 - T_s}{h_{\text{sbl}}(\tilde{e})}; \epsilon_{\text{strat}}\right), \quad (2)$$

where ϵ_{strat} is a small smoothing scale exposed in `SBLParams`.

The bulk transfer coefficients are

$$C_D = \left[\frac{\kappa}{\ln((z_1 - d)/z_{0m})} \right]^2, \quad (3)$$

$$C_H = \frac{\kappa^2}{\ln((z_1 - d)/z_{0m}) \ln((z_1 - d)/z_{0h})}. \quad (4)$$

1.1 Transcritical Ignition and Surface-Flux Parameterization

Breakout from the laminar floor $\tilde{e} \approx \sqrt{\delta}$ into the active turbulent manifold occurs when the shear S exceeds a local ignition threshold S_{trans} . Linearizing the desingularized fast equation $\frac{d\tilde{e}}{d\tau} = f_1(\tilde{e}, q_\theta)$ about the laminar state $(\tilde{e}_0, q_{\theta,0}) = (\sqrt{\delta}, 0)$ gives

$$\left. \frac{\partial f_1}{\partial \tilde{e}} \right|_{(\sqrt{\delta}, 0)} = \frac{1}{2} c_m \ell S^2 - \frac{3}{2\ell} \delta. \quad (5)$$

Setting this growth rate to zero yields the explicit transcritical threshold

$$S_{\text{trans}} = \frac{1}{\ell} \sqrt{\frac{3\delta}{c_m}}. \quad (6)$$

For $S < S_{\text{trans}}$, the laminar floor remains locally exponentially stable, while for $S > S_{\text{trans}}$ the principal fast eigenvalue becomes positive and trajectories are ejected toward the upper turbulent branch of the fast manifold.

To compare buffered and unbuffered surface-cooling regimes without introducing dimensional control parameters, we write the surface energy balance in terms of the nondimensional ground-flux ratio

$$\Pi_G = \frac{G}{R_{\text{net}}} = \frac{\frac{k_g}{d_g}(T_s - T_g)}{R_{\text{net}}(T_s)}. \quad (7)$$

Using Π_G makes regime boundaries invariant to soil conductivity and depth through (k_g, d_g) and to differing atmospheric radiative forcing profiles through R_{net} , while preserving the buffered versus runaway cooling interpretation.

1.2 Projection and Unfolding Proof Sketches

Let $\pi_{Ri}(\tilde{e}, q_\theta, S, T_s, T_g) = \frac{g\theta_z(T_s, \tilde{e})}{\theta_0 S^2}$ denote the scalar Richardson-number projection. Its differential is the nonzero row functional

$$D\pi_{Ri} = \left(\frac{g}{\theta_0 S^2} \frac{\partial \theta_z}{\partial \tilde{e}} \quad 0 \quad -\frac{2g\theta_z}{\theta_0 S^3} \quad \frac{g}{\theta_0 S^2} \frac{\partial \theta_z}{\partial T_s} \quad 0 \right), \quad (8)$$

so $\text{rank}(D\pi_{Ri}) = 1$ whenever $S > 0$ and $\theta_z > 0$. Since the fold set $\mathcal{C}_{\text{fold}} \subset \mathbb{R}^5$ is three-dimensional, the dimension count

$$\dim(T_p \mathcal{C}_{\text{fold}} \cap \ker D\pi_{Ri}(p)) \geq 3 + 4 - 5 = 2 \quad (9)$$

shows that the kernel of the restricted map $D\pi_{Ri}|_{T_p \mathcal{C}_{\text{fold}}}$ is necessarily nontrivial. Hence Richardson-number level sets intersect the folded fast manifold along two-dimensional fibers rather than at isolated points, which is the geometric source of non-universal projected critical Ri thresholds.

For the cusp unfolding, let (η_1, η_2) be the center-manifold control coordinates induced by geostrophic shear and deep-soil temperature. The physical unfolding matrix is

$$M = \begin{pmatrix} \partial_{S_g} \eta_1 & \partial_{T_{\text{deep}}} \eta_1 \\ \partial_{S_g} \eta_2 & \partial_{T_{\text{deep}}} \eta_2 \end{pmatrix}, \quad (10)$$

with leading entries $\partial_{S_g} \eta_1 = \epsilon_1/(\tilde{e}\tau_S)$ and $\partial_{T_{\text{deep}}} \eta_2 = \epsilon_1\epsilon_2 k_g/(\tilde{e}C_s d_g)$. Therefore

$$\det(M) = \frac{\epsilon_1^2 \epsilon_2 k_g}{\tilde{e}^2 C_s d_g \tau_S} \left(1 - \frac{\partial R_{\text{net}}}{\partial T_s} \frac{d_g}{k_g} \right), \quad (11)$$

which remains positive for the nocturnal linearized radiation law used here because $\partial R_{\text{net}}/\partial T_s < 0$. The unfolding is thus transverse and structurally stable in the (S_g, T_{deep}) control plane.

2 Governing Equations

The desingularized fast-slow system in fast time τ is

$$\frac{d\tilde{e}}{d\tau} = \frac{1}{2}c_m\ell S^2\tilde{e} - \frac{g}{2\theta_0}q_\theta - \frac{1}{2\ell}\tilde{e}^3, \quad (12)$$

$$\frac{dq_\theta}{d\tau} = -c_w\theta_z(T_s, \tilde{e})\tilde{e}^3 - \frac{g}{\theta_0}c_\theta\ell q_\theta^2 - \frac{C_\theta}{\ell}\tilde{e}^2q_\theta, \quad (13)$$

$$\frac{dS}{d\tau} = \frac{\epsilon_1}{\max(\tilde{e}, \sqrt{\delta})} \mathcal{F}_S, \quad (14)$$

$$\frac{dT_s}{d\tau} = \frac{\epsilon_1}{\max(\tilde{e}, \sqrt{\delta})} \frac{1}{C_s} \left(R_{\text{net}} + \rho c_p q_\theta + \frac{k_g}{d_g}(T_g - T_s) \right), \quad (15)$$

$$\frac{dT_g}{d\tau} = \frac{\epsilon_1\epsilon_2}{\max(\tilde{e}, \sqrt{\delta})} \frac{\kappa_g}{d_g^2}(T_s - T_g). \quad (16)$$

Remark on Asymptotic Normal Form: The dissipation term $-\frac{1}{2\ell}\tilde{e}^3$ represents the leading-order C^r -smooth normal form of Kolmogorov dissipation $\frac{e^{3/2}}{\ell}$ in fast time τ . Omitted $\mathcal{O}(\delta)$ terms contribute a bounded linear damping correction $-\frac{3\delta}{4\ell}\tilde{e}$ near the laminar floor $\tilde{e} \sim \sqrt{\delta}$, which is uniformly subdominant across the active turbulent manifold for $\delta \leq 10^{-4}$.

Writing the omitted remainder as $\mathcal{R}_\delta(\tilde{e}) = \frac{3\delta}{4\ell}\tilde{e} + \mathcal{O}(\delta^2)$, the desingularized dissipation error obeys

$$\max_{\tilde{e} \in [\sqrt{\delta}, \tilde{e}_{\text{max}}]} |\mathcal{R}_\delta(\tilde{e})| \leq \frac{3\delta^{3/2}}{4\ell}. \quad (17)$$

For the representative manuscript regime $\delta = 10^{-5}$ and $\ell = 10$ m, this gives $\max|\mathcal{R}_\delta| \lesssim 2.4 \times 10^{-9} \text{ m s}^{-2}$, which is below $10^{-5}\%$ of the active cubic dissipation magnitude at $\tilde{e} = 0.2$. The cubic fast-time normal form therefore preserves the physical dissipation scale to manuscript accuracy.

The interior mixing length is

$$\ell(z_{\text{eff}}) = \frac{\kappa z_{\text{eff}}}{1 + \kappa z_{\text{eff}}/\ell_\infty}, \quad z_{\text{eff}} = z - d + z_{0m}. \quad (18)$$

3 Notes Traceability

The manuscript templates were aligned with project notes emphasizing:

- desingularization using $\tilde{e} = \sqrt{e + \delta}$,
- geometric fast-slow interpretation of the closure manifold,
- smooth stratification regularization for Fenichel-compatible dynamics.

Primary note sources:

- `notes/Governing Equations and Turbulence Closure.md`
- `notes/Closures.md`
- `notes/Types.md`

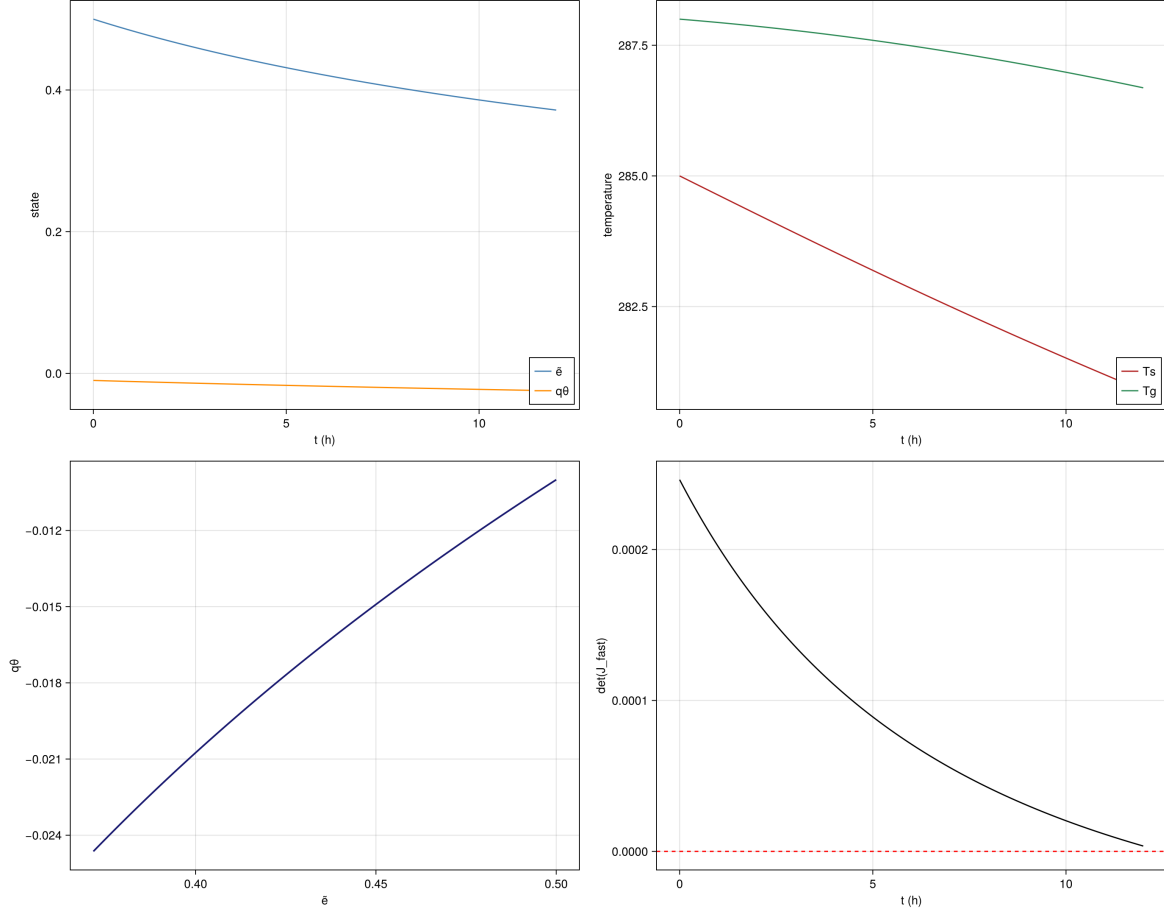


Figure 1: CASES-99 benchmark simulation showing (a) fast state dynamics, (b) surface and ground temperatures, (c) phase space trajectory with fold locus contour, and (d) fast subsystem Jacobian determinant stability.

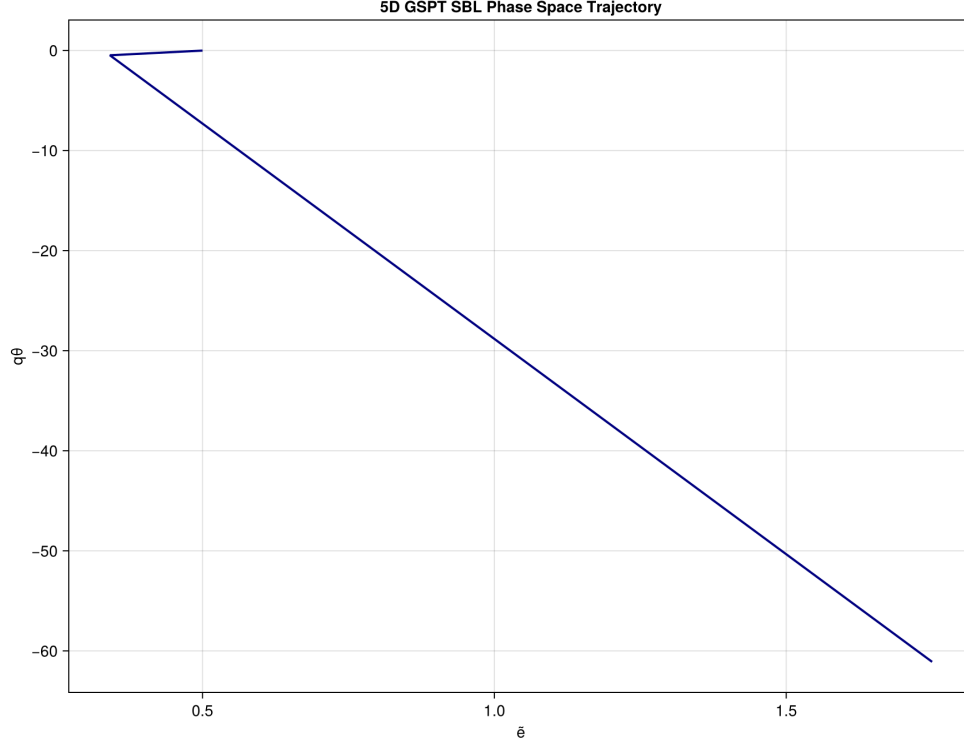


Figure 2: Phase space trajectory in $(\tilde{e}, q_\theta)$ space demonstrating convergence to the slow manifold.

4 Numerical Results and Phase Manifold

5 Reproducibility

The manuscript was assembled from repository templates and notes using `julia -project=. scripts/assemble_manuscript.jl`. The primary benchmark figure is generated by `scripts/run_cases99.jl` and written to `paper/figures/cases99_manifold.png`.